

Operations Workflow Management System (OWMS)

Complete Database Schema Documentation

ENUM: task_status

Value	Description
PENDING	Task is awaiting action
IN_PROGRESS	Task is currently being processed
REJECTED	Task was rejected at some step
COMPLETED	Task successfully finished all steps
ESCALATED	Task exceeded SLA and was escalated

ENUM: approval_decision

Value	Description
APPROVED	Step approved
REJECTED	Step rejected

roles

Column	Type	Description
id	UUID (PK)	Primary Key, Default uuid_generate_v4()
name	VARCHAR(50)	Unique Role Name (ADMIN, MANAGER, OPERATOR)
created_at	TIMESTAMP	Default CURRENT_TIMESTAMP

users

Column	Type	Description
id	UUID (PK)	Primary Key

Column	Type	Description
name	VARCHAR(100)	User Full Name
email	VARCHAR(150)	Unique Email
password_hash	TEXT	Encrypted Password
role_id	UUID (FK)	References roles(id)
is_active	BOOLEAN	Account Active Status
created_at	TIMESTAMP	Account Creation Time

workflows

Column	Type	Description
id	UUID (PK)	Primary Key
name	VARCHAR(150)	Workflow Name
description	TEXT	Workflow Description
created_by	UUID (FK)	References users(id)
is_active	BOOLEAN	Workflow Active Status
created_at	TIMESTAMP	Creation Time

workflow_steps

Column	Type	Description
id	UUID (PK)	Primary Key
workflow_id	UUID (FK)	References workflows(id)
step_order	INT	Step Sequence Order
role_required	UUID (FK)	References roles(id)
sla_hours	INT	Allowed SLA Hours
created_at	TIMESTAMP	Creation Time

tasks

Column	Type	Description
id	UUID (PK)	Primary Key
workflow_id	UUID (FK)	References workflows(id)

Column	Type	Description
current_step	INT	Current Step Number
status	task_status ENUM	Task Status
created_by	UUID (FK)	References users(id)
assigned_to	UUID (FK)	References users(id)
priority_level	INT	1=High, 2=Medium, 3=Low
rejection_count	INT	Number of rejections
escalation_count	INT	Number of escalations
priority_score	FLOAT	Computed Intelligent Score
started_at	TIMESTAMP	Start Time
due_at	TIMESTAMP	SLA Due Time
completed_at	TIMESTAMP	Completion Time
created_at	TIMESTAMP	Creation Time

approvals

Column	Type	Description
id	UUID (PK)	Primary Key
task_id	UUID (FK)	References tasks(id)
step_id	UUID (FK)	References workflow_steps(id)
approved_by	UUID (FK)	References users(id)
decision	approval_decision ENUM	Approval Decision
comments	TEXT	Optional Comments
decided_at	TIMESTAMP	Decision Timestamp

escalations

Column	Type	Description
id	UUID (PK)	Primary Key
task_id	UUID (FK)	References tasks(id)
escalated_at	TIMESTAMP	Escalation Time
reason	TEXT	Escalation Reason
resolved	BOOLEAN	Escalation Resolved Status

audit_logs

Column	Type	Description
id	UUID (PK)	Primary Key
user_id	UUID (FK)	References users(id)
action	VARCHAR(100)	Action Performed
entity_type	VARCHAR(50)	Entity Type
entity_id	UUID	Entity Identifier
metadata	JSONB	Additional Structured Data
created_at	TIMESTAMP	Log Timestamp